Grade 10 – Term 1: C8 Classroom Activity

Name: _____ Date: _____

Goal: Compare decision criteria, their information needs, priorities, and limitations. All entries are profits; larger is better. Use supplied summaries unless asked to update them. Report all ties. Give brief explanations; use your notebook for extra working.

Guided Problems

1. Delivery expansion. Profits are in thousand US dollars.

	Market payoffs			Supplied summary			
Plan	Strong	Stable	Weak	Row max.	Row min.	Max. regret	EMV
Probability	.20	.60	.20				
A_1: Large fleet	100	20	-20	100	-20	55	28
A_2: Flexible fleet	75	50	10	75	10	25	47
A_3: Current fleet	40	40	35	40	35	60	39
A_4: Partners	60	65	20	65	20	40	55

(a) Complete the selections. Maximax: highest row maximum; Maximin: highest row minimum; Minimax Regret: smallest maximum regret; EMV: highest expected profit.

Criterion	Alternative(s)	Deciding quantity and value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

(b) Complete: only _____ needs probabilities. Regret compares plans within each _____. Match these priorities to criteria: greatest upside; best worst outcome; least maximum missed opportunity; highest expected money.

(c) Compare A_2 and A_4: give a numerical reason for each choice and one limitation of each criterion used. Does EMV 55 guarantee profit 55?

2. Festival catering. Profits are in thousand US dollars.

	High	Low	Row max.	Row min.	Max. regret	EMV
Probability	.30	.70				
A_1: Large contract	80	10	80	10	25	31
A_2: Small contract	45	35	45	35	35	38

(a) Record all four recommendations and deciding values, in the same order as Problem 1.

(b) Low attendance: A_1 earns 10 and forgoes $35 - 10 =$ _____. Explain why this regret is not a cash loss. Does the regret choice protect the highest minimum profit?

(c) New probabilities are $(.60, .40)$. Complete $EMV(A_1) = .60(80) + .40(10) =$ _____ and $EMV(A_2) = .60(45) + .40(35) =$ _____. State the new EMV choice and explain why the other three choices stay unchanged.

Standard Problems

For each problem, report all four recommendations with their deciding quantities and values. Justify comparisons numerically; mark EMV unavailable if necessary.

3. Retail plans: agreement and ties. Profits are in thousand CNY.

	Strong	Normal	Weak	Row max.	Row min.	Max. regret	EMV
Probability	.25	.50	.25				
A_1	60	45	30	60	30	0	45
A_2	50	40	20	50	20	10	37.5
A_3	60	35	30	60	30	10	40

(a) Report recommendations, including ties. (b) Compare A_1 with each other plan state by state. Explain why two criteria still permit ties and why agreement does not make the criteria interchangeable.

Criterion	Alternative(s)	Deciding quantity and value
Maximax		
Maximin		
Minimax Regret		
EMV/Bayes		

4. Urban farm: subsidy and constraint. Profits are in thousand US dollars.

	Strong	Stable	Weak	Row max.	Row min.	Max. regret	EMV
Probability	.50	.30	.20				
A_1: Specialty	80	40	-10	80	-10	45	50
A_2: Balanced	60	55	20	60	20	20	50.5
A_3: Staple	40	45	35	45	35	40	40.5

(a) Report unrestricted recommendations. (b) Profit must be at least 20 in every state. A subsidy adds an integer s, $0 \le s \le 20$, only to A_2's weak-state payoff. Find the smallest s making Maximin strictly prefer A_2 to A_3. Identify eligible plans. At that subsidy, calculate updated maxima, minima, regrets, and EMVs using eligible plans only, then report all four restricted recommendations.

5. Coastal ferry: no forecast. Profits are in thousand US dollars. No reliable state probabilities are available.

	High	Normal	Low	Row max.	Row min.	Max. regret
A_1: Two ferries	100	35	-25	100	-25	55
A_2: One ferry	75	55	10	75	10	25
A_3: Lease capacity	55	48	30	55	30	45
A_4: Current fleet	30	30	30	30	30	70

(a) Report recommendations and explain why equal probabilities cannot be assumed. (b) Choose a criterion for an operator prioritizing missed opportunities. Contrast it with worst-case protection, giving numerical evidence and a limitation of each. (c) A later forecast gives $(.20, .30, .50)$. Calculate all EMVs and select a plan. Explain EMV's risk orientation and one limitation.

Solutions

1. Delivery expansion

(a) Compare each supplied summary column:

i. Maximax: $\max(100, 75, 40, 65) = 100$: A_1, row maximum 100.
ii. Maximin: $\max(-20, 10, 35, 20) = 35$: A_3, row minimum 35.
iii. Minimax Regret: $\min(55, 25, 60, 40) = 25$: A_2, maximum regret 25.
iv. EMV/Bayes: $\max(28, 47, 39, 55) = 55$: A_4, EMV 55. No ties occur.

(b) The blanks are EMV/Bayes and state. Greatest upside: optimistic Maximax; best worst outcome: conservative Maximin; least maximum missed opportunity: Minimax Regret; highest expected money: risk-neutral EMV/Bayes. Maximax and Maximin need row payoffs; regret needs the full payoff table for comparisons within states. These three require no probabilities. EMV also requires nonnegative state probabilities summing to one, here $.20 + .60 + .20 = 1$.

(c) Minimax Regret favors A_2: maximum regret $25 < 40$ for A_4. It limits missed opportunities but does not protect the highest minimum profit: A_2 can earn only 10. EMV favors A_4: $.20(60) + .60(65) + .20(20) = 12 + 39 + 4 = 55 > 47$ for A_2. It depends on the forecast and does not guarantee 55: actual profits are 60, 65, or 20. Maximax's optimistic choice ignores A_1's possible loss of 20; Maximin protects 35 with A_3 but ignores the upside of 100 available from A_1.

2. Festival catering

(a) Maximax: A_1, row maximum $80 > 45$. Maximin: A_2, row minimum $35 > 10$. Minimax Regret: A_1, maximum regret $25 < 35$. EMV: A_2, EMV $38 > 31$. There are no ties.

(b) Low-state best profit is 35, so regret is $35 - 10 = 25$. The large contract earns positive profit 10; regret 25 is forgone profit, not a cash loss. State bests $(80, 35)$ give regret rows $(0, 25)$ and $(35, 0)$. Minimax Regret chooses A_1 even though its minimum profit is 10, below A_2's 35; it does not maximize the minimum profit.

(c) $EMV(A_1) = 48 + 4 = 52$; $EMV(A_2) = 27 + 14 = 41$. The probabilities sum to one. EMV now selects A_1 (52), instead of A_2 (38 under the original forecast). Payoffs and regrets are unchanged: Maximax stays A_1 (80), Maximin stays A_2 (35), and Minimax Regret stays A_1 (25).

3. Retail plans

(a) Maximax compares $(60, 50, 60)$: A_1, A_3 tie at row maximum 60. Maximin compares $(30, 20, 30)$: A_1, A_3 tie at row minimum 30. Minimax Regret compares $(0, 10, 10)$: A_1 wins with maximum regret 0. EMV compares $(45, 37.5, 40)$: A_1 wins with EMV 45.

(b) A_1 beats A_2 in every state: $60 > 50$, $45 > 40$, $30 > 20$. Against A_3, it is equal in strong and weak states and better in normal: $60 = 60$, $45 > 35$, $30 = 30$. Maximax and Maximin ignore the normal-state difference because the row maxima and minima tie. State bests $(60, 45, 30)$ give regret rows $(0, 0, 0)$, $(10, 5, 10)$, and $(0, 10, 0)$, distinguishing A_1 from A_3. EMVs are

$$EMV(A_1) = .25(60) + .50(45) + .25(30) = 15 + 22.5 + 7.5 = 45,$$
$$EMV(A_2) = .25(50) + .50(40) + .25(20) = 12.5 + 20 + 5 = 37.5,$$
$$EMV(A_3) = .25(60) + .50(35) + .25(30) = 15 + 17.5 + 7.5 = 40.$$

Agreement does not make the criteria interchangeable. They prioritize optimistic upside, conservative minimum, limited missed opportunities, and risk-neutral expected money, respectively. Only EMV needs probabilities. Maximax and Maximin miss the normal-state advantage here; regret ignores likelihoods; EMV depends on probability estimates and does not guarantee its average (45 versus a possible 30 for A_1).

4. Urban farm

(a) Comparing supplied columns gives Maximax: A_1 (row maximum 80); Maximin: A_3 (row minimum 35); Minimax Regret: A_2 (maximum regret 20); EMV: A_2 (50.5). There are no ties.

(b) A_1 is ineligible since $-10 < 20$. For $0 \le s \le 20$, A_2 has minimum $20 + s \le 40 < 55 < 60$, and A_3 has minimum 35; both are eligible. Strict Maximin preference requires

$$20 + s > 35 \iff s > 15.$$

The smallest integer is $s = 16$, or \$16,000. At $s = 15$ there is only a tie. At $s = 16$, eligible payoff rows are $(60, 55, 36)$ and $(40, 45, 35)$. Their maxima are 60 and 45, and minima 36 and 35. State bests are $(60, 55, 36)$. Subtract each eligible payoff from its state best:

$$r(A_2) = (0, 0, 0), \qquad r(A_3) = (20, 10, 1).$$

Maximum regrets are 0 and 20. The updated EMVs are

$$EMV(A_2) = .50(60) + .30(55) + .20(36) = 30 + 16.5 + 7.2 = 53.7,$$
$$EMV(A_3) = .50(40) + .30(45) + .20(35) = 20 + 13.5 + 7 = 40.5.$$

All four restricted criteria select A_2: Maximax with row maximum 60; Maximin with row minimum 36; Minimax Regret with maximum regret 0; EMV with 53.7. Maximax changes because A_1 is excluded; Maximin changes because the subsidy raises A_2's minimum above 35. Regrets must be recomputed after excluding A_1.

5. Coastal ferry

(a) Maximax: A_1, row maximum 100. Maximin: A_3, A_4, tied at row minimum 30. Minimax Regret: A_2, maximum regret 25. EMV and its overall recommendation are unavailable without state probabilities. Although A_4 has constant payoff 30, the other plans' expected values cannot be compared without a distribution. Equal probabilities add an unsupported assumption of equally likely states.

(b) Minimax Regret fits the missed-opportunity priority: A_2 limits maximum regret to 25, versus 45 for A_3 and 70 for A_4. It compares payoffs within states without probabilities, but can recommend a minimum profit of only 10. Maximin protects a minimum of 30 with A_3 or A_4, also without probabilities. It ignores upside and likelihoods: A_3 can earn 55 while A_4 never exceeds 30, yet they tie.

(c) All probabilities are nonnegative and $.20 + .30 + .50 = 1$.

$$EMV(A_1) = .20(100) + .30(35) + .50(-25) = 20 + 10.5 - 12.5 = 18,$$
$$EMV(A_2) = .20(75) + .30(55) + .50(10) = 15 + 16.5 + 5 = 36.5,$$
$$EMV(A_3) = .20(55) + .30(48) + .50(30) = 11 + 14.4 + 15 = 40.4,$$
$$EMV(A_4) = .20(30) + .30(30) + .50(30) = 6 + 9 + 15 = 30.$$

EMV selects A_3 (40.4), with no tie. It is risk-neutral about money and depends on reliable probabilities; 40.4 is not guaranteed since A_3 may earn 30. The three probability-free recommendations stay unchanged because payoffs are unchanged. EMV and Maximin agree on A_3 for different reasons.